

Automated Plants Watering System

AYAN PAHWA

During summers, most people are too lazy to water the potted plants on their rooftop gardens every day. Explained in this section is a simple and exciting plant watering system that you can build yourself in just a few hours. It is an Arduino based automatic plant water-feeder system that uses a soil moisture sensor. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the automated plant watering system is shown in Fig.

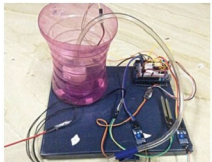


Fig. 1: Author's prototype

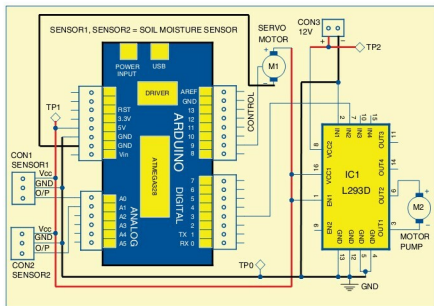


Fig. 2: Circuit diagram of the automated plants watering system

2. The circuit comprises an Arduino UNO board, a soil moisture sensor, a servo motor, a 12V water pump and an L293D (IC1) motor driver IC to run the water pump.

You can power the Arduino board using a 7V to 12V wall wart or plug-in adaptor or solar panel. You need a separate 12V battery or power supply or solar panel for the pump motor.

Soil moisture sensor. Two types of soil moisture sensors are available in the market—contact and non-contact sensors. A contact soil sensor (as shown in Fig. 3) is used in this project because it has to check soil moisture to measure the electrical conductivity.

The moisture sensor provides an analogue output, which can easily be interfaced with Arduino. In this project, two sensors can be connected to analogue pins, A0 and A1, of the Arduino board. Each sensor has four pins (Vcc, Gnd, Ao and Do) available for interfacing with the Arduino board. Here, digital output pin (Do) is not used. The water pump and servo motor are controlled by Arduino connected to digital pins 3 and 9, respectively. That is, the



servo motor signal control pin is connected to pin 9 of the Arduino board.

The program in the Arduino reads the moisture value from the sensor every 20 seconds. If the value reaches the threshold value, the program does the following three things:

1. It moves the servo motor horn, along with the water pipe fixed on it, toward potted plant, whose moisture level is less than the predetermined/threshold level.

2. It starts the motor pump to supply water to the plant for a fixed period of time and then stops the water pump (refer Fig. 4).

3. It brings back the servo motor horn to its initial position.

Software program

The program is written in Arduino programming language. The code is well commented and is easy to understand. Compile the autowatering.ino code and upload it to the micro-

controller, using Arduino IDE version 1.

The sensor will calibrate by itself once it is kept in the soil

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	12V

PARTS LIST

Semiconductor:

- IC1 - L293D motor driver
- Soil moisture sensor
- Arduino UNO board

Miscellaneous:

- M1 - Servo motor
- M2 - 12V DC motor pump
- CON1, CON2 - 3-pin connector
- CON3 - 2-pin connector
- Water container
- Small flexible water pipe
- 12V battery
- 7-12V power adaptor